

A4.4.1 Ethical implications of machine learning



A 4.4.1 Ethical implications of machine learning

A4.4.1 discusses ethical implications of ML in real-world scenarios: accountability, fairness, bias (especially from training data), consent, environmental/societal impact, privacy, security, and transparency, including online communication issues like harassment.

Introduction

As we've seen, machine learning gives us the power to build powerful tools that can predict, classify, and create. But with this power comes a profound responsibility. Just because we *can* build something, does that mean we *should*? This section explores the complex ethical minefield of machine learning and why, as a computer scientist, thinking about ethics is just as important as writing code.

The Core Ethical Issues in Machine Learning

When we deploy ML models in the real world, they can have a huge impact on people's lives. We must consider a wide range of ethical issues to ensure these systems are fair, safe, and beneficial.

1. Bias and Algorithmic Fairness

- **What is it? Bias** occurs when an ML model produces results that are systematically prejudiced or unfair towards certain groups. This almost always happens because the data used to train the model reflects historical or societal biases. **Algorithmic Fairness** is the goal of ensuring a model's outcomes are equitable and not discriminatory.
- **Real-World Example:** An AI recruiting tool was trained on a company's hiring data from the past 10 years. Because the company had historically hired more men for technical roles, the model learned to penalise CVs that contained the word "women's" (e.g., "captain of the women's chess club"). It was replicating the bias in the data, unfairly disadvantaging female candidates.

2. Accountability

- **What is it?** This is the question of who is responsible when an AI system makes a mistake or causes harm. Is it the programmer who wrote the code? The company that deployed the system? The user who operated it? Or the AI itself?
- **Real-World Example:** If a self-driving car causes an accident, who is at fault? There is currently no clear legal or ethical answer, and this is a major challenge for the industry.

3. Transparency

- **What is it?** This refers to how well we can explain the reasoning behind a model's decision. Many powerful models, especially deep neural networks, are considered "black boxes" because their decision-making process is so complex that it's impossible for a human to fully understand.
- **Real-World Example:** A bank uses a deep learning model to approve or deny loans. If a person is denied, and the bank cannot explain *why* (due to the model's complexity), the customer has no way to appeal the decision or know how to improve their chances in the future.

4. Privacy and Consent

- **What is it?** ML models require vast amounts of data to train, and this data is often personal. This raises questions about how our data is being collected, stored, and used. Did we give informed **consent** for our data to be used to train a commercial AI?
- **Real-World Example:** Social media companies use your photos, posts, and interactions to train their recommendation and advertising algorithms. While you agree to terms of service, it's debatable whether users fully understand the extent to which their personal data is being used.

5. Security

- **What is it?** This concerns the vulnerability of ML models to attack. Models can be tricked or manipulated. For example, "adversarial attacks" can make tiny, human-imperceptible changes to an image that cause a model to completely misclassify it.
- **Real-World Example:** Researchers have shown they can place a few small stickers on a "STOP" sign that make a self-driving car's vision system classify it as a "Speed Limit 45" sign, which could have catastrophic consequences.

6. Environmental Impact

- **What is it?** Training large-scale machine learning models, like the ones that power ChatGPT or image generators, requires an enormous amount of computational power. This is done in massive data centres that consume vast amounts of electricity, contributing to a significant carbon footprint.
- **Real-World Example:** It has been estimated that training a single large AI model can emit as much carbon as five cars over their entire lifetimes.

7. Societal Impact

- **What is it?** This is the broad consideration of how the widespread adoption of AI affects society as a whole. This includes issues like job displacement (will AI automate jobs?), economic inequality, and how AI changes the way we interact with each other.

A Deeper Look: The Challenge of Biased Training Data



The single biggest source of ethical problems in ML is biased data. An algorithm is only as good as the data it learns from. If the data is a reflection of a flawed and biased world, the model will learn and often amplify those biases. This is the principle of "Garbage In, Garbage Out."

Example: Predictive Policing. If a police department trains an AI on historical arrest data to predict where crime is likely to occur, the model will likely send more officers to neighborhoods that have been historically over-policed, regardless of the actual crime rate. This creates a feedback loop: more police presence leads to more arrests, which "proves" to the model that its prediction was correct, leading to even more bias.

Case Study: The Ethics of ML in Online Communication



Social media and content platforms are driven by ML algorithms designed to maximise one thing: **engagement**. This single-minded goal can have serious negative consequences.

Misinformation and Filter Bubbles: Algorithms on platforms like YouTube or TikTok learn what keeps you watching. They often discover that sensational, extreme, and emotionally charged content (including misinformation) is highly engaging. By constantly recommending this type of content, they can push users into "filter bubbles" or "echo chambers," where their existing beliefs are constantly reinforced and they are not exposed to differing viewpoints, leading to societal polarisation.

Online Harassment: While ML can be used to detect and block online abuse, it also presents challenges. Algorithms can struggle with context, sarcasm, and cultural nuance, leading to both censorship of legitimate speech and failure to stop sophisticated harassment.

Anonymity and Privacy: ML algorithms are incredibly powerful at finding patterns. They can sometimes **de-anonymise** individuals by combining multiple, seemingly anonymous pieces of data to re-identify a person, creating a significant privacy risk.

Flashcards

Use this Quizlet to learn all the key terms in this section.

Match



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode ▼

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

A company develops an AI hiring tool by training it on their last 20 years of employee data. The tool is found to systematically rank male candidates higher than female candidates. This is a classic example of:

A. ☐ An adversarial attack

B. ☒ Algorithmic bias from training data

- C. ☐ Poor model security
- D. ☐ A lack of accountability

2

The question "If a self-driving car has to choose between hitting a pedestrian or swerving and harming its passenger, what should it do?" is a problem of:

- A. ☐ Privacy
- B. ☐ Transparency
- C. ☒ Accountability and ethical decision-making
- D. ☐ Environmental impact

3

A deep learning model that is so complex its creators cannot fully explain why it made a particular decision is often referred to as a:

- A. ☐ Supervised model
- B. ☐ Biased model
- C. ☐ General-purpose model
- D. ☒ "Black box" model

4

What is the most significant environmental concern associated with training large-scale AI models?

- A. ☐ The physical waste from old computer parts.
- B. ☒ The high electricity consumption of data centres.
- C. ☐ The use of rare earth metals in the hardware.

D. ☐ The noise pollution from cooling fans.

5

A "filter bubble" in online communication is an ethical concern because it can lead to:

A. ☐ A decrease in user engagement.

B. ☒ Societal polarisation by isolating users from different viewpoints.

C. ☐ A higher risk of the website being hacked.

D. ☐ An increase in online privacy.

6

The ethical principle that deals with determining who is at fault when an AI system fails is:

A. ☐ Transparency

B. ☒ Accountability

C. ☐ Fairness

D. ☐ Consent

7

Why must ethical guidelines for technology be continually reassessed?

A. ☐ Because the hardware becomes outdated.

B. ☐ Because data scientists forget the old guidelines.

C. ☒ Because technology advances rapidly, creating new capabilities and unforeseen ethical issues.

D. ☐ Because old guidelines expire after a few years.



8

Putting small, specially designed stickers on a STOP sign to make an AI misclassify it is an example of:

- A. ☐ A privacy breach
- B. ☐ A filter bubble
- C. ☒ An adversarial attack (a security issue)
- D. ☐ An environmental impact

9

The primary reason an ML model might be biased is that:

- A. ☐ It was not trained for long enough.
- B. ☐ The algorithm itself was programmed to be unfair.
- C. ☐ The hardware it runs on is too slow.
- D. ☒ The data it was trained on contained human or societal biases.

10

Which of the following is a key ethical concern related to the use of personal photos from social media to train a commercial facial recognition model?

- A. ☒ Privacy and Consent
- B. ☐ Environmental Impact
- C. ☐ Algorithmic Fairness
- D. ☐ Accountability

All materials on this website are for the exclusive use of teachers and students at subscribing schools for the period of their subscription. Any unauthorised copying or posting of materials on other websites is an infringement of our copyright and could result in your account being blocked and legal action being taken against you.

 **Report a problem**